LYLE OKOTH

AI PRODUCT ENGINEER - FULL-STACK BACKEND AND AI FEATURES

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PROFESSIONAL SUMMARY

AI Product Engineer who ships speech and generative products end to end - research, model, backend, billing and interface - across **6 launched products** in **6+ years**. Built **Voxa** solo from brief to live service, including accounts, **3 billing paths** and a creator studio.

Domain-driven **FastAPI** services with **5 bounded contexts**, Google OAuth, **M-Pesa, Stripe and card billing**, storage portable across **R2 / S3 / Azure**, and SSE streaming with **first audio chunk under 400ms**.

TECHNICAL SKILLS

- **Backend:** Python, FastAPI, REST, SSE and WebSockets, Celery + Redis job queues, domain-driven design, ports and adapters, PostgreSQL, retry-safe async pipelines
- **Front-end and UX:** JavaScript, React, Next.js, real-time streaming UIs, waveform and transcript views, Figma flows, onboarding design
- **AI integration:** Hugging Face Transformers, Whisper, VITS and flow-matching TTS, Stable Diffusion with LoRA, LLM prompt pipelines, inference APIs, GPU worker queues
- **Auth, billing and storage:** Google OAuth, session handling, M-Pesa API, Stripe and card billing, Cloudflare R2, S3, Azure Blob behind one swappable interface
- **Delivery and quality:** Docker, GitHub Actions, AWS, Playwright E2E tests, logging and tracing, latency and cost tuning

WORK EXPERIENCE

Machine Learning Researcher | SautiFlow Labs — 2026 - Present

Diffusion and flow-matching speech synthesis for languages with almost no public data.

- Trained a flow-matching TTS model covering **8 Kenyan languages** at **0.2s real-time factor** on a single consumer GPU, bringing multilingual synthesis inside interactive latency budgets.
- Cut **character error rate from 21% to 9%** on in-house Swahili and Sheng evaluation sets through fine-tuning, corpus curation and accent-aware text normalisation.

AI Product Developer | Freelance — 2026 - Present

End-to-end ownership on a small number of products: research, flow design, model, launch, pricing, operation.

- Built **Voxa** alone from brief to live product: speech-skit generation, account system, **3 billing paths** and a studio interface for creators.
- Made infrastructure portable by design - storage swappable across local disk, **R2, S3 and Azure** with **no application code changes**.
- Modelled **Sheng and code-switched Kenyan dialogue** inside the product flow rather than bolting localisation on afterwards.
- Owned the full path: Figma flows, **FastAPI / PostgreSQL / Redis / Celery** backend, React interface and Playwright end-to-end tests.

Machine Learning Engineer | SautiFlow Labs — 2022 - 2026

Owned the path from checkpoint to service: pipelines, inference APIs, evaluation harnesses, cost control.

- Shipped the SemaScribe speaker-diarization pipeline, processing **50,000 hours of meeting audio at 12% diarization error rate**.
- Served speech inference on FastAPI + Celery + Redis at **8,000 requests/day at 180ms p95**, using ONNX Runtime, dynamic batching and GPU worker pools.

- Cut **training spend by 40%** with mixed precision and preemptible GPU capacity at **equal model quality**, verified on frozen evaluation sets.

Software and Data Engineer | SautiFlow Labs **2022 - 2025**

Two sides of one job: move data reliably, and ship a product people can actually use.

- Built SnapSavanna, a dataset and labeling platform hosting **18 African-language datasets and 40,000 labeled samples** with content-hash versioning.
- Designed domain-driven FastAPI services split into **5 bounded contexts** with ports and adapters at the edges, keeping model code out of the product path.
- Shipped SuraYa and EchoBuni with Google OAuth, **M-Pesa and card billing**, and swappable object storage across **R2, S3 and Azure**.
- Delivered server-sent events streaming for progressive audio generation with **first audio chunk under 400ms**.
- Built Airflow orchestration over Pandas, Parquet and DuckDB for ingestion, de-duplication and split generation.

SELECTED PROJECTS

SautiFlow - AI Audio Skit Studio

Script to finished skit: prompt expansion into speaker-labelled scenes, multilingual TTS per character, automatic timing, line-level re-takes, 15/30/60s export presets.

Tech: Python, FastAPI, PyTorch, VITS, Whisper, FFmpeg, Celery, Redis, PostgreSQL, Docker, AWS S3

PlusOne - AI Personal Photographer

Event galleries from 4-6 selfies: per-person LoRA adapters, ControlNet composition, ArcFace identity gate, quality scoring and Real-ESRGAN print-resolution upscaling.

Tech: PyTorch, Stable Diffusion, LoRA, ControlNet, ArcFace, Real-ESRGAN, FastAPI, Celery, Stripe

Vocalis - Voice Design and Speech Studio

Flow-matching TTS studio: brand voice design, cloning from short consented samples, 24/48 kHz masters, consent checks and audible watermarking, GPU inference queue with streamed output.

Tech: PyTorch, torchaudio, VITS, Whisper, ONNX Runtime, FastAPI, Redis, Celery, React, Kubernetes

EDUCATION

BSc Computer Security and Forensics | Jaramogi Oginga Odinga University of Science and Technology (JOOUST) **[20XX - 20XX]**

ML, signal processing and security coursework; final-year project on audio watermarking for corpus provenance.

Mechatronics Engineering | Jomo Kenyatta University of Agriculture and Technology (JKUAT) **[20XX - 20XX]**

Fourier analysis, sampling theory, filter design and real-time embedded constraints - the grounding behind latency work.

OPEN SOURCE, WRITING AND COMMUNITY

- **Open ML releases:** 4 models and 3 datasets on Hugging Face, **12,000+ downloads**; model cards and training recipes per release.
- **Writing and tooling:** speech AI articles on Medium (@lyle-okoth); Python libraries for the YouTube and Google Drive APIs on PyPI.
- **Community:** OMDENA volunteer (2024); co-founder, SautiFlow Labs (2022).